

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

EIGENVALUES & EIGENVECTORS

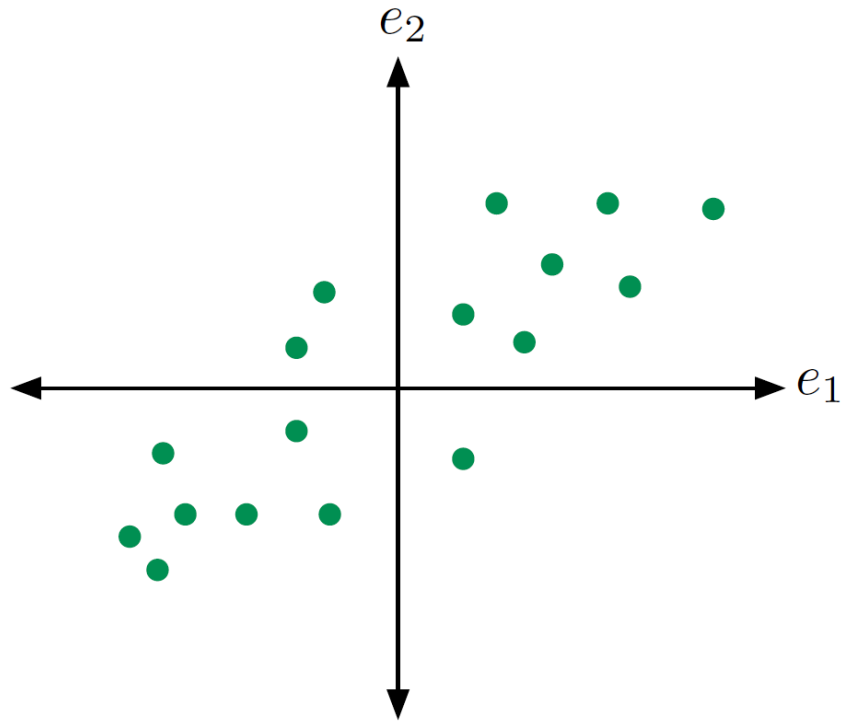
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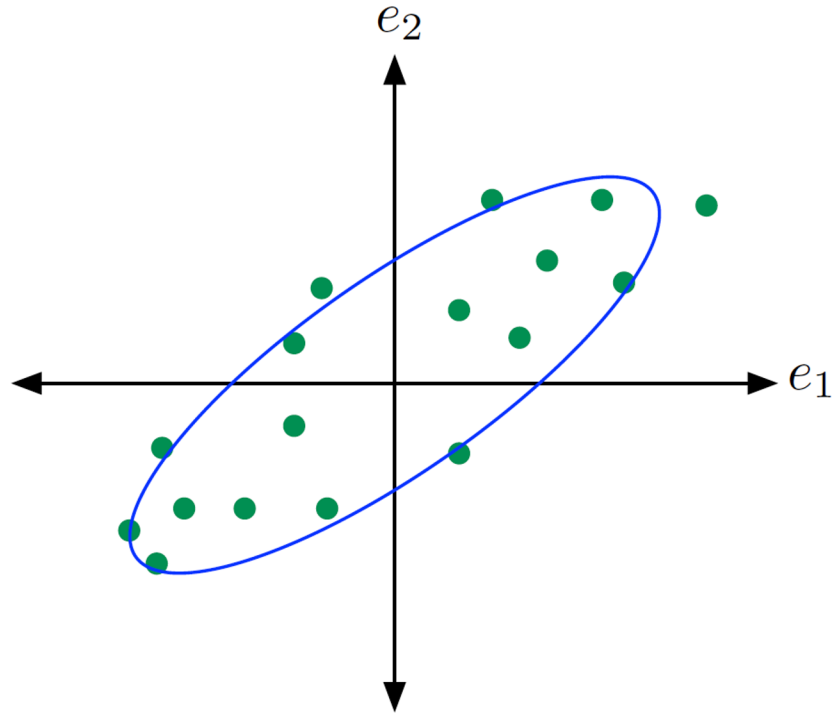
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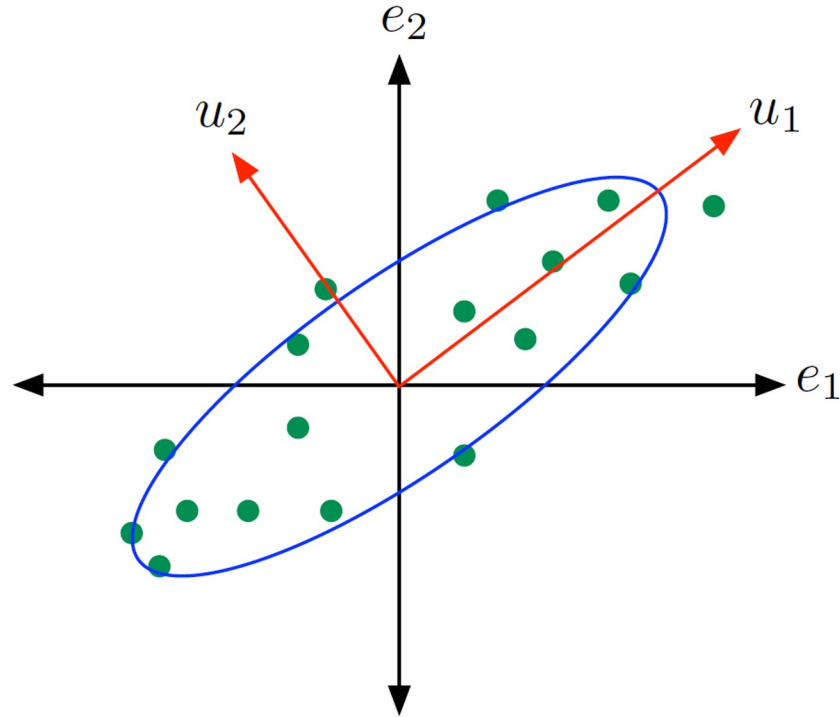
Linear Algebra: Review of Eigendecomposition



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Eigenvector & Eigenvalue: Definition

Let M be any $d \times d$ matrix.

- M defines a linear function, $x \mapsto Mx$. This maps $\mathbb{R}^d$ to $\mathbb{R}^d$.

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Let M be any $d \times d$ matrix.

- M defines a linear function, $x \mapsto Mx$. This maps $\mathbb{R}^d$ to $\mathbb{R}^d$.
- We say $u \in \mathbb{R}^d$ is an **eigenvector** of M if

$$Mu = \lambda u$$

for some scaling constant λ . This λ is the **eigenvalue** associated with u .

- Key point: M **maps eigenvector u onto the same direction.**

Question: What are the eigenvectors and eigenvalues of:

$$M = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 10 \end{pmatrix} ?$$

Eigenvectors of a Real Symmetric Matrix

Fact: Let M be any real symmetric $d \times d$ matrix. Then M has

- d eigenvalues $\lambda_1, \dots, \lambda_d$
- corresponding eigenvectors $u_1, \dots, u_d \in \mathbb{R}^d$ that are orthonormal

Can think of $u_1, \dots, u_d$ as the axes of the natural coordinate system for M .

Example

$$M = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} \text{ has eigenvectors } u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

- 1 Are these orthonormal?
- 2 What are the corresponding eigenvalues?